

Sean Casey Hall

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Aron Culotta

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Fine-Tuning Transformer Models for Question Answering: BERT and its Variants

In this project, I fine-tuned and evaluated four transformer-based models in the context of question answering (QA) tasks: BERT, ALBERT, RoBERTa, and DistilBERT. These models were trained on Stanford's SQuAD 2.0 dataset and evaluated on Stanford's SQuAD 1.1, SQuAD 2.0, AddSent, and AddOneSent. The performance of these models are measured in Exact Match (EM) and F1 scores. The transformer-based models are benchmarked against a simple Bag-of-Words baseline model. A Lite BERT (ALBERT) showed the strongest overall performance and demonstrated a serious adversarial robustness. RoBERTa performed the strongest on SQuAD 2.0 and DistilBERT offered a very comparable efficiency at a fraction of the size. All models were vulnerable to the adversarial datasets of AddSent and AddOneSent; ALBERT being the most resilient. The purpose of this project is to demonstrate the trade-offs between model size, accuracy, and robustness.

Despite the overall success of these models, adversarial datasets have proven that these models can be sensitive to minor input perturbations. This poses a challenge for real-world deployment where ambiguity and noise are common. This project attempts to address this challenge by evaluating the robustness of many well known transformer architecture on standard and adversarial datasets.

There has been extensive research on BERT and its variants. BERT (Devlin et al., 2018) introduced a revolutionary pretraining approach that proved to be highly successful. Through

their use of Masked Language Modeling (MLM) and Next Sentence Prediction (NSP), the group achieved state-of-the-art results on many NLP tasks, including QA on SQuAD. My project uses BERT as the baseline model for comparison, specifically on the task on QA. RoBERTa (Lan et al., 2020) built upon BERT by removing NSP, implementing dynamic masking, and training on more data in larger batch sizes. This improved performance on SQuAD without any architectural changes. In my project, I implement RoBERTa to assess the impact of these new pretraining steps on adversarial QA tasks. ALBERT (Lan et al., 2020) introduced cross-layer parameter sharing and factorized embeddings. This significantly reduced the number of parameters, while maintaining performance and even improving when trained for a long enough time. This model replaced NSP with Sentence Order Prediction in an attempt to improve inter-sentence coherence. My result showed that ALBERT is the most robust out of all the models tested, consistent with the findings in its original paper. DistilBERT (Sanh et al., 2019) focused on efficiency. This was done by reducing the size of BERT through knowledge distillation. In this paper, the model retained 97% of BERT's performance on GLUE benchmarks despite being significantly faster and lighter. In my project, DistilBERT demonstrates the performance-efficiency tradeoff, especially when faced with adversarial inputs. This paper (Jia & Liang, 2017) introduces two adversarial datasets, AddSent and AddOneSent, designed to test the brittleness of QA models. These datasets highlight how high-performing models can be easily tripped up by syntactically plausible but semantically irrelevant distractors. My project evaluates all four models using these two adversarial benchmarks to measure the QA model's robustness.

For this project, I have 3 goals. The first is to evaluate the accuracy and efficiency of BERT, RoBERTa, ALBERT, and DistilBERT on QA tasks. The second is to test how these models perform on adversarial datasets, simulating real-world noise and ambiguity. The third is

to identify trade-offs between model size, computational complexity, and model robustness, helping inform model deployment decisions.

My approach focuses on fine-tuning BERT, RoBERTa, ALBERT, and DistilBERT for extractive question answering. I used Hugging Face's Transformers and Datasets libraries to implement, train, and evaluate these models. All of the models were fine-tuned on the SQuAD 2.0 dataset, which includes both answerable and unanswerable questions. The dataset tests the model, not only on its ability to accurately extract answers from a passage, but also to recognize when no answer exists.

The transformer models used in this project are BERT and its variants. BERT was developed by Google and improves natural language processing by developing an understanding of context in text. BERT stands for Bidirectional Encoder Representations from Transformers which learns context by masking 15% of all tokens in its context window and predicts the masked token. It is a multi-head, multi-layer transformer that attends to both all the tokens before and after the masked token. Its variants explored in this project include: A lite BERT (ALBERT), A Robust BERT (RoBERTa), and a distilled version of BERT (DistilBERT). BERT base has 110M parameters and is pre-trained using static Masked Language Modeling (MLM) where 15% of all input tokens are masked before training and Next Sentence Prediction (NSP) where the model learns to predict whether one sentence logically follows another. BERT provides a strong baseline for QA tasks. Among its variants, RoBERTa uses the same architecture as BERT, but its training is further optimized and is slightly larger (125M parameters). RoBERTa is not trained on NSP and uses dynamic masking where 15% of tokens are randomly masked during training, and changes every time the model iterates through the training data, leading to a more robust model. RoBERTa is also trained for longer, and on a larger corpora than BERT. ALBERT is significantly

smaller than BERT, with only 12M parameters and achieves this parameter reduction by factorized embeddings and cross-layer parameter sharing. It also replaces NSP in its training with Sentence Order Prediction (SOP) where the model learns to determine if two consecutive segments from the same text are in the correct order or have been swapped. This leads to better model coherence which plays a huge role when evaluating adversarial datasets. Lastly, DistilBERT comes in at 66M parameters. It is a distilled version of BERT and is trained to replicate BERT's logits (pre softmax outputs). Instead of discovering the complex relationship between words and their meanings, it attempts to copy BERT's hidden states. It has fewer layers, no NSP, making it faster with slightly reduced performance.

All models use their respective pretrained tokenizers. In this project, each model uses a maximum sequence length of 384 and a document stride of 128 for splitting long contexts. The optimizer used is AdamW with learning rate warm-up and linear decay schedule. Each model ran for multiple epochs, with early stopping based on validation metrics.

To ensure these models are worth their complexity, I compared them with a simple Bag-of-Words benchmark. For this model, I used TF-IDF to vectorize the context and the question. Then I retrieved the sentence with the highest cosine similarity to the question. Then, for dealing with Squad 2.0 with unanswerable questions, I implemented a confidence threshold which could produce “no answer”. This model serves the purpose of acting as a baseline for the transformer models, highlighting the gap in capability between classical and multi-layered neural approaches.

All models were trained on SQuAD 2.0 and evaluated on SQuAD 1.1, SQuAD 2.0, AddSent, and AddOneSent. SQuAD 1.1 is Stanford's dataset consisting of 100,000 QA pairs with guaranteed answers found in the text. SQuAD 2.0 got released later and includes 150,000

QA pairs with the addition of unanswerable questions, forcing the model to admit when no answer exists within the passage. AddSent is an adversarial dataset made to look identical to SQuAD's 1.1 QA pairs but with a distractor sentence that contains part of the question added. AddOneSent is another adversarial dataset that inserts a completely random sentence with no semantic relevance to the question. These adversarial datasets are crucial for testing resilience to minor perturbations and testing real-world noise and ambiguity.

The models were evaluated using two standard QA metrics: Exact Match (EM) and F1 Score. EM is a binary score (0 or 1) that measures whether the predicted answer matches the ground truth exactly. F1 score is the harmonic mean between precision and recall, measuring overlapping tokens between predicted and gold answers. These two metrics provide complementary views of accuracy as EM is strict while F1 gives partial credit.

Many of the models performed well on SQuAD 1.1 and 2.0. I noticed an overall drop off in performance when evaluated on AddOneSent and an even larger performance degradation on AddSent. This is intuitive as an inserted sentence with remnants of the question in it will more likely trip up the model than a completely random sentence (AddOneSent). When considering the adversarial datasets, ALBERT performed the best with RoBERTa not far behind. BERT placed third followed by DistilBERT, followed by the significantly worse BoW model. For the SQuAD datasets, ALBERT performed the best, doing barely better than RoBERTa. BERT was not far behind in third, and DistilBERT came in last again. The results can be seen in the table below.

Model	SQuAD 1.1 (EM/F1)	SQuAD 2.0 (EM/F1)	AddSent (EM/F1)	AddOneSent (EM/F1)
ALBERT	76.39 / 83.22	75.67 / 79.26	55.84 / 60.52	64.13 / 69.70
RoBERTa	71.16 / 82.93	75.67 / 81.58	50.90 / 59.74	59.15 / 69.31
BERT	70.58 / 77.71	71.72 / 75.53	50.65 / 56.22	57.97 / 64.02
DistilBERT	60.70 / 67.99	66.27 / 70.08	40.87 / 46.05	47.68 / 53.89
BoW Benchmark	0.25 / 19.00	0.93 / 10.74	0.08 / 16.59	0.17 / 17.57

ALBERT consistently outperformed the other models across all datasets despite being the smallest in size (12M). I believe this is strongly correlated with its SOP pre-training task and parameter sharing. This likely contributed to better cross-sentence reasoning and generalization, allowing the model to accurately disregard unrelated sentences. RoBERTa performed strongly on SQuAD 2.0 suggesting that its size and large-scale training helped identify unanswerable questions. DistilBERT was efficient, but most vulnerable to adversarial noise, showing significant performance degradation, especially on AddSent. Finally, the BoW model served as a useful lower bound. As anticipated, the simple, shallow model performed extremely poor on all fronts.

After the evaluation was done, I analyzed the types of questions each model got wrong on every dataset. I noticed that ALBERT, our strongest model, struggled with punctuation, potentially implying a limitation with parameter sharing. RoBERTa's errors were almost always because the model's output was too verbose. The model tended to include too much context and over-predicted answer spans. DistilBERT was very easily misled by adversarial sentences with question-like tokens. AddSent, as made evident by the model's scores, was more damaging to

model performance than AddOneSent for all models. This is because neural network models can better distinguish an unrelated sentence from one, more closely aligned with the question.

This project demonstrates the practical trade-offs and vulnerabilities of transformer-based models in QA tasks. All models achieved strong performances on standard benchmarks, and exhibited a degraded performance when faced with adversarial inputs. Most surprisingly, ALBERT, despite its small parameter count, showed the best robustness and overall accuracy across all standard and adversarial datasets. This highlights the value of smart architectural choices such as parameter sharing and sentence coherence modeling. RoBERTa showed a competitive performance, especially on SQuAD 2.0, validating the importance of large models pretraining on large datasets. Future work should include next-generation models like ELECTRA which uses replaced-token detection, east of masking, for efficient pretraining. Another model, DeBERTa, separates position embeddings and content information, allowing the model to have a deeper contextual understanding. Additionally, data augmentation strategies could help mitigate performance drops on noisy, ambiguous inputs such as the adversarial datasets used. Ultimately, this project has shown that architectural efficiency and robustness do not need to be mutually exclusive. Lightweight models like ALBERT can compete and outperform heavier models if designed and trained thoughtfully. As transformer models become more integrated into real-world systems, it is crucial to understand these trade-offs for reliable and secure NLP applications.

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